

Daily Geometry Drill #3 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Relative Positions · Radical Axis by subtraction · Common Chords · Orthogonal Circles · Tangency to Both Axes.

Section A Rapid Recognition

1. The circle $x^2 + y^2 - 12x - 16y + 75 = 0$ is tangent to the circle $x^2 + y^2 = 25$. Write down the number of common tangents of the two circles.

Answer: 3

Method: $x^2 + y^2 - 12x - 16y + 75 = (x - 6)^2 + (y - 8)^2 - 100 + 75 = 0$, so centre (6, 8), radius $\sqrt{36 + 64 - 75} = 5$. Both circles have radius 5; $d = 10 = r_1 + r_2$ — externally tangent — 3 common tangents.

Investigate further: The bookkeeping: $d > r_1 + r_2 \Rightarrow 4$, $d = r_1 + r_2 \Rightarrow 3$, $|r_1 - r_2| < d < r_1 + r_2 \Rightarrow 2$, $d = |r_1 - r_2| \Rightarrow 1$, $d < |r_1 - r_2| \Rightarrow 0$. **“Tangent” means exactly one common point, giving exactly 3 tangents, not 2.**

2. The circles with centre (0,0) radius 5 and centre (13,0) radius 3 are disjoint. Write down the number of common tangents of the two circles.

Answer: 4

Method: $d = 13$, $r_1 + r_2 = 8$; since $d > r_1 + r_2$ the circles are fully disjoint and there are 4 common tangents.

Investigate further: Disjoint circles admit 2 direct (external) tangents plus 2 transverse (internal) tangents. **Once the circles overlap, the two transverse tangents disappear.**

3. Write down the radical axis of the circles $x^2 + y^2 - 6x + 2y + 1 = 0$ and $x^2 + y^2 - 2x - 4y + 13 = 0$.

Answer: $2x - 3y + 6 = 0$

Method: At the radical axis powers are equal, so subtract: $(-6x + 2y + 1) - (-2x - 4y + 13) = 0$, i.e. $-4x + 6y - 12 = 0$, i.e. $2x - 3y + 6 = 0$. The $x^2 + y^2$ terms cancel in one stroke.

Investigate further: Check the answer by symmetry: $x = \frac{3}{4}y - \frac{3}{2}$ plug-in balances both powers automatically. **A single-line subtraction beats solving any intersection; the radical axis exists even for disjoint circles.**

4. Write down whether the circles $x^2 + y^2 = 9$ and $x^2 + y^2 + 6x + 8y + 9 = 0$ are orthogonal (yes or no).

Answer: yes

Method: $x^2 + y^2 + 6x + 8y + 9$ has centre $(-3, -4)$ and $r^2 = 9 + 16 - 9 = 16$, so $r = 4$. $d = \sqrt{3^2 + 4^2} = 5$. Orthogonal iff $d^2 = r_1^2 + r_2^2$: $25 = 9 + 16$ ✓. Yes.

Investigate further: Equivalently with coefficients: $g_1 = f_1 = 0$, $g_2 = 3$, $f_2 = 4$ and $2g_1g_2 + 2f_1f_2 = 0$ must equal $c_1 + c_2 = (-9) + (9) = 0$. **Orthogonality is about the radii forming a right triangle with the centre distance.**

5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 10x + 5 = 0$ intersect in two points. Find the length of their common chord.

Answer: 8

Method: Subtract: $-25 = -10x + 5 \Rightarrow x = 3$, the radical axis. Distance of $x = 3$ from the origin is 3, radius 5, so half-chord $= \sqrt{25 - 9} = 4$, chord $= 8$.

Investigate further: The second circle has centre $(5, 0)$ and radius $\sqrt{20}$; the two circles agree on the two end-points of this chord. **Identify the radical axis first — the chord sits on it — then apply the half-chord formula.**

6. Two circles have radii 3 and 5 and their centres are 4 units apart. Write down the number of common tangents.

Answer: 2

Method: $|r_1 - r_2| = 2 < d = 4 < r_1 + r_2 = 8$ — intersecting circles — 2 common tangents.

Investigate further: Two crossing circles always have exactly 2 common tangents. **The trap is counting the two intersection points as “tangents”.**

7. The circle $x^2 + y^2 - 10x - 10y + k = 0$ is tangent to both coordinate axes. Find k .

Answer: 25

Method: Centre $(5, 5)$. Tangent to both axes forces the radius to equal each coordinate: $r = 5$. Then $r^2 = 25 = 5^2 + 5^2 - k$, so $k = 50 - 25 = 25$.

Investigate further: A circle tangent to both axes has centre (r, r) , $(-r, r)$, $(r, -r)$ or $(-r, -r)$ and equation $(x \mp r)^2 + (y \mp r)^2 = r^2$, constant r^2 . **The constant is r^2 , not $2r^2$.**

8. The circle $x^2 + y^2 + 4x - 2y + C = 0$ is orthogonal to the circle $x^2 + y^2 - 4x + 4y - 1 = 0$. Find C .

Answer: -11

Method: $g_1 = 2$, $f_1 = -1$; $g_2 = -2$, $f_2 = 2$, $c_2 = -1$. Orthogonality: $2g_1g_2 + 2f_1f_2 = c_1 + c_2 \Rightarrow 2(2)(-2) + 2(-1)(2) = C - 1 \Rightarrow -8 - 4 = C - 1 \Rightarrow C = -11$.

Investigate further: Geometric check: first circle centre $(-2, 1)$ with $r_1^2 = C$? No — $r_1^2 = 4 + 1 - C = 5 - C$; second centre $(2, -2)$ with $r_2^2 = 9$. $d^2 = 25 = 9 + 5 - C \Rightarrow C = -11$ ✓. **Both forms of the test mirror each other.**

9. How many common tangents do the circles $x^2 + y^2 - 4x - 6y + 12 = 0$ and $x^2 + y^2 + 2x - 10y + 25 = 0$ have?

Answer: 4

Method: First: centre $(2, 3)$, $r^2 = 4 + 9 - 12 = 1$. Second: centre $(-1, 5)$, $r^2 = 1 + 25 - 25 = 1$.

$d = \sqrt{(3)^2 + (2)^2} = \sqrt{13} > 2 = r_1 + r_2$ — disjoint unit circles — 4 common tangents.

Investigate further: Completing the square (Worksheet #2 skill) locates the centres; only then compare d with $r_1 + r_2$. **Worksheets #2's centre-and-radius reading is today's first move.**

10. Two circles have radii 7 and 3 and exactly 3 common tangents. Find the distance between their centres.

Answer: 10

Method: Exactly 3 common tangents means the circles are externally tangent, so $d = r_1 + r_2 = 7 + 3 = 10$.

Investigate further: Keep the five-case table on one line. **If there were 2 tangents you would instead need $|3 - 7| < d < 10$, which leaves a whole range, not a single value.**

Section B Manipulation Drills

1. How many common tangents do the circles $x^2 + y^2 = 9$ and $x^2 + y^2 - 4x - 6y + 9 = 0$ have?

- A) 0
- B) 1
- C) 2
- D) 3
- E) 4

Answer: C) 2

Method: $x^2 + y^2 - 4x - 6y + 9 = 0$: centre $(2, 3)$, $r^2 = 4 + 9 - 9 = 4$. The other has centre $(0, 0)$ radius 3. $d = \sqrt{13}$; $|3 - 2| = 1 < \sqrt{13} < 5$ — intersecting — 2 common tangents.

Investigate further: $x^2 + y^2 = 9$ is radius 3, so the second circle is strictly inside the first's 5-radius relation. **Do not stop after finding the centres; compare d against both $r_1 + r_2$ and $|r_1 - r_2|$.**

2. The radical axis of the circles $x^2 + y^2 + ax + by + c = 0$ and $x^2 + y^2 + 6x - 4y + 10 = 0$ is $2x - y + 3 = 0$. What is $a + b$?

- A) 3
- B) -3
- C) 13
- D) -13

Answer: A) 3

Method: Subtract: $(a - 6)x + (b + 4)y + (c - 10) = 0$. Compare with $2x - y + 3 = 0$: $a - 6 = 2 \Rightarrow a = 8$; $b + 4 = -1 \Rightarrow b = -5$; $c - 10 = 3 \Rightarrow c = 13$. So $a + b = 3$.

Investigate further: Each coefficient matches with the *same* scale factor (here 1); change the factor and all three coefficient equations change together. **The y -coefficient sign is the usual slip — $b + 4 = -1$ gives $b = -5$, not -3 .**

3. The circle $x^2 + y^2 - 4x + 2y - 5 = 0$ is orthogonal to $x^2 + y^2 + 6x - 2y + k = 0$. Find k .

- A) -9

- B) 9
C) -5
D) 14

Answer: A) -9

Method: $g_1 = -2, f_1 = 1, c_1 = -5; g_2 = 3, f_2 = -1, c_2 = k$. $2g_1g_2 + 2f_1f_2 = 2(-2)(3) + 2(1)(-1) = -14 = c_1 + c_2 = -5 + k \Rightarrow k = -9$.

Investigate further: Orthogonal circles cut at right angles: the radii to a common point are perpendicular. **Never pair g with c or f with g — the pairing is $g_1g_2 + f_1f_2$ against $\frac{c_1+c_2}{2}$.**

4. A circle is tangent to both coordinate axes in the first quadrant and passes through the point (1, 8). Find the smaller possible radius.

Answer: 5

Method: Centre (r, r) , radius r : $(1 - r)^2 + (8 - r)^2 = r^2 \Rightarrow r^2 - 18r + 65 = 0 \Rightarrow (r - 5)(r - 13) = 0$. Smaller radius 5.

Investigate further: There really are two such circles — always state which root the question wants. **Both roots are positive, so the phrasing “smaller” is essential.**

5. Two circles of equal radius 5 have centres (0, 0) and (6, 0). Their radical axis has equation?

- A) $x = 3$
B) $x = 6$
C) $y = 3$
D) $y = 0$

Answer: A) $x = 3$

Method: Equal radii: subtract $(x^2 + y^2 - 25) = ((x - 6)^2 + y^2 - 25) \Rightarrow -25 = -12x + 36 - 25 \Rightarrow x = 3$.

Investigate further: For equal circles the radical axis is the perpendicular bisector of the centre segment. **A quick sanity check: power at (3, 0) is $9 - 25 = -16$ for both sides.**

6. Two circles of radii 4 and 6 have centres 9 units apart. How many common tangents do they have?

- A) 0
B) 1
C) 2
D) 3
E) 4

Answer: C) 2

Method: $|r_1 - r_2| = 2 < d = 9 < 10 = r_1 + r_2$: strict overlap without containment — 2 common tangents.

Investigate further: Compare d to 10 (sum) and 2 (difference) before counting. 9 is **dangerously close to 10; only exactly 10 would give 3.**

7. The circle $x^2 + y^2 = 4$ is orthogonal to $x^2 + y^2 + 8x - 6y + C = 0$. Find C .

- A) 4
B) -4
C) 25
D) 2

Answer: A) 4

Method: Second circle: centre $(-4, 3)$, $r_2^2 = 25 - C$. $d^2 = 16 + 9 = 25$. Orthogonal: $d^2 = r_1^2 + r_2^2 = 4 + (25 - C) = 29 - C \Rightarrow C = 4$.

Investigate further: Check with $g_1 = f_1 = 0$: the coefficient test gives $0 = -4 + C \Rightarrow C = 4 \checkmark$.
Both routes agree — use whichever you trust.

8. The circles $x^2 + y^2 = 36$ and $(x - 6)^2 + y^2 = 36$ intersect. Find the length of their common chord.

Answer: $6\sqrt{3}$

Method: Equal radii 6 on centres 6 apart: radical axis is the perpendicular bisector $x = 3$. Half-chord $= \sqrt{36 - 9} = 3\sqrt{3}$, chord $= 6\sqrt{3}$.

Investigate further: The two circles overlap in a lens whose extreme width is $6\sqrt{3}$. **Do not add the two radii; the chord lives on the radical axis, not at the intersection of centre-lines.**

9. A circle is tangent to both coordinate axes in the first quadrant and passes through the point $(4, 8)$. Its possible radii are?

- A) 4 only
B) 20 only
C) 4 and 20
D) 16

Answer: C) 4 and 20

Method: Centre (r, r) : $(4 - r)^2 + (8 - r)^2 = r^2 \Rightarrow r^2 - 24r + 80 = 0 \Rightarrow (r - 4)(r - 20) = 0$. Both 4 and 20.

Investigate further: Verify: radius 4 circle touches both axes near the origin; radius 20 circle is far out and still reaches $(4, 8)$. **A quadratic in r can give two genuine circles — list both.**

10. The point $P = (1, t)$ lies on the radical axis $3x - 4y + 5 = 0$ of two circles. Find t .

Answer: 2

Method: Substitute: $3(1) - 4t + 5 = 0 \Rightarrow 8 - 4t = 0 \Rightarrow t = 2$.

Investigate further: The radical axis is exactly the set of points of equal power — a line. **Put P into the line, not into either circle equation.**

Section C Substitution & Structure

1. The circle with centre $(2, -1)$ and radius 5 and the circle $x^2 + y^2 - 12x - 4y + 24 = 0$ intersect. How do they meet?
- A) at two points
 - B) at exactly one point (tangent)
 - C) not at all; one contains the other
 - D) not at all; disjoint exteriors

Answer: A) at two points

Method: Second circle: centre $(6, 2)$, $r^2 = 36 + 4 - 24 = 16$. $d = \sqrt{16 + 9} = 5$; $|5 - 4| = 1 < 5 < 9$ — two points.

Investigate further: The common centres distance and the two radii are exactly the three sides of a valid triangle ($1 < 5 < 9$). **A geometric “triangle inequality” check is faster than solving anything.**

2. Two circles of radii 3 and 4 are orthogonal. What is the distance between their centres?
- A) 5
 - B) 7
 - C) 1
 - D) $\sqrt{7}$

Answer: A) 5

Method: Orthogonal iff $d^2 = r_1^2 + r_2^2$: $d^2 = 9 + 16 = 25 \Rightarrow d = 5$.

Investigate further: 3-4-5 right triangle in disguise. **Option C claims $|4 - 3| = 1$; that is the containment boundary, not orthogonality.**

3. Two circles of radii 3 and 8 have centres 13 units apart. The direct common tangent segment between the two circles has what length?
- A) 12
 - B) 5
 - C) 8
 - D) $\sqrt{104}$

Answer: A) 12

Method: Direct tangent segment $= \sqrt{d^2 - (r_2 - r_1)^2} = \sqrt{13^2 - 5^2} = 12$.

Investigate further: Drop perpendiculars from both centres to the common tangent: a right triangle of legs 12 and 5, hypotenuse 13. **Option D arises from squaring $r_2 + r_1 = 11$ into the bracket instead of 5.**

4. For the same two circles (radii 3 and 8, centres 13 apart), the transverse common tangent segment has what length?

- A) $4\sqrt{3}$
- B) $6\sqrt{3}$
- C) 12
- D) $2\sqrt{6}$

Answer: A) $4\sqrt{3}$

Method: Transverse tangent segment = $\sqrt{d^2 - (r_1 + r_2)^2} = \sqrt{169 - 121} = \sqrt{48} = 4\sqrt{3}$.

Investigate further: The transverse tangent crosses the gap between the circles, so the bracket holds the *sum* of radii 11. **Switching the bracket from 5 to 11 is exactly the C3-to-C4 distinction.**

5. Which of the following is the radical axis of the circles $x^2 + y^2 - 2x - 4y - 8 = 0$ and $x^2 + y^2 + 4x - 2y + 3 = 0$?
- A) $6x + 2y + 11 = 0$
 - B) $6x - 2y + 11 = 0$
 - C) $6x + 2y - 11 = 0$
 - D) $2x + 6y + 11 = 0$

Answer: A) $6x + 2y + 11 = 0$

Method: Subtract: $(-2x - 4y - 8) = (4x - 2y + 3) \Rightarrow -6x - 2y - 11 = 0 \Rightarrow 6x + 2y + 11 = 0$.

Investigate further: The x and y coefficients swap roles between C and D — keep track of which circle owned which term. **Check your result by substituting any point that satisfies both.**

6. How many circles tangent to both coordinate axes pass through the point $(9, 2)$?
- A) 0
 - B) 1
 - C) 2
 - D) 3

Answer: C) 2

Method: Centre (r, r) : $(9 - r)^2 + (2 - r)^2 = r^2 \Rightarrow 2r^2 - 22r + 85 = 0 \Rightarrow r^2 - 22r + 85 = 0 \Rightarrow (r - 5)(r - 17) = 0$. Two positive radii, so two circles.

Investigate further: Both $r = 5$ and $r = 17$ satisfy the geometry; the quadratic produced two genuine circles. **Whether a question wants the count (2) or the radii (5 and 17) — answer exactly what is asked.**

7. The circles $x^2 + y^2 = 169$ and $x^2 + y^2 - 6x + 8y - 119 = 0$ meet in two points. What is the length of their common chord?
- A) 12
 - B) 16
 - C) 24

D) 26

Answer: C) 24

Method: Second circle: centre $(3, -4)$, $r^2 = 9 + 16 + 119 = 144$, $r = 12$. Subtract: $-169 = -6x + 8y - 119 \Rightarrow 6x - 8y - 50 = 0 \Rightarrow 3x - 4y - 25 = 0$. Distance from origin $= |-25|/5 = 5$. Half-chord $= \sqrt{169 - 25} = 12$, chord $= 24$.

Investigate further: The 13-12-5 Pythagorean triple runs the whole question. **Sanity-check: the chord length 24 is visibly less than the diameter $2 \cdot 13 = 26$ (option D).**

8. The circles $x^2 + y^2 + 4x - 1 = 0$, $x^2 + y^2 - 8y + 11 = 0$ and $x^2 + y^2 + 2x - 12y + 13 = 0$ have pairwise radical axes that all pass through a single point. This radical centre has coordinates?

A) $(1, 1)$ B) $(1, -1)$ C) $(3, 1)$ D) $(1, 2)$ **Answer:** A) $(1, 1)$

Method: Radical axis of the first two: $(4x - 1) - (-8y + 11) = 0 \Rightarrow x + 2y - 3 = 0$. Of the second two: $(-8y + 11) - (2x - 12y + 13) = 0 \Rightarrow x - 2y + 1 = 0$. Solve: $x + 2y = 3$, $x - 2y = -1 \Rightarrow x = 1, y = 1$.

Investigate further: Verify the power: at $(1, 1)$, circle 1 gives $1 + 1 + 4 - 1 = 5$; circle 2 gives $1 + 1 - 8 + 11 = 5$; circle 3 gives $1 + 1 + 2 - 12 + 13 = 5$ ✓. **Finding two radical axes meets at the radical centre; the third axis must pass through it.**

Section D Challenge Ramp

1. A point P moves so that its power with respect to the circle $x^2 + y^2 = 1$ is equal to its power with respect to the circle $(x - 6)^2 + y^2 = 4$. As P moves it traces out?

A) the line $x = \frac{11}{4}$ B) the circle $x^2 + y^2 = 1$

C) a pair of isolated points

D) the y -axis**Answer:** A) the line $x = \frac{11}{4}$

Method: Equal powers ARE the radical axis: $x^2 + y^2 - 1 = (x - 6)^2 + y^2 - 4 = x^2 + y^2 - 12x + 32 \Rightarrow 12x = 33 \Rightarrow x = \frac{11}{4}$. A vertical line.

Investigate further: The locus of equal powers is always a straight line, even when the circles are disjoint — here $d = 6 > 3 = r_1 + r_2$. **Do not picture a “middle circle”; the radical axis is a line, not a curve.**

2. The circles $x^2 + y^2 = 100$ and $x^2 + y^2 - 6x - 8y - 40 = 0$ intersect in two points. What is the length of their common chord?

- A) 8
- B) 16
- C) 12
- D) $6\sqrt{3}$

Answer: B) 16

Method: Second circle: centre $(3, 4)$, $r^2 = 9 + 16 + 40 = 65$. Subtract: $-100 = -6x - 8y - 40 \Rightarrow 3x + 4y = 30$. Distance to origin $= 30/5 = 6$. Half-chord $= \sqrt{100 - 36} = 8$, chord $= 16$.

Investigate further: Twice over: distance from $(3, 4)$ to $3x + 4y - 30$ is $|9 + 16 - 30|/5 = 1$, giving half-chord $\sqrt{65 - 1} = 8$ ✓. **Both centres agree the chord is 16 — a cheap consistency check.**

3. The circle $x^2 + y^2 - 6x - 8y + C = 0$ is orthogonal to the circle with centre $(2, -1)$ and radius 5. Find C .

- A) 24
- B) 26
- C) 25
- D) -24

Answer: A) 24

Method: Circle 1: centre $(3, 4)$, $r_1^2 = 25 - C$. Circle 2: centre $(2, -1)$, $r_2 = 5$. $d^2 = (3 - 2)^2 + (4 + 1)^2 = 1 + 25 = 26$. Orthogonality: $26 = 25 - C + 25 \Rightarrow C = 24$.

Investigate further: The partner circle $(x - 2)^2 + (y + 1)^2 = 25$ is Worksheet #2's A3 — orthogonality lets today's sheet lean on yesterday's circle. **Keep $r_1^2 = 25 - C$, not $C - 25$.**

4. Two circles of radii 3 and 4 are orthogonal. What is the length of their common chord?

- A) $\frac{24}{5}$
- B) $\frac{12}{5}$
- C) 6
- D) $\frac{48}{5}$

Answer: A) $\frac{24}{5}$

Method: $d^2 = r_1^2 + r_2^2 = 9 + 16 = 25$, $d = 5$. Place centres $(0, 0)$ and $(5, 0)$; the radical axis solves $-9 = -10x + 9 \Rightarrow x = \frac{9}{5}$. Half-chord of the radius-3 circle: $\sqrt{9 - (\frac{9}{5})^2} = \sqrt{\frac{144}{25}} = \frac{12}{5}$, chord $= \frac{24}{5}$.

Investigate further: Work the 3-4-5 system: the radical axis sits $\frac{9}{5}$ from the small circle's centre, and the chord is the small circle's own chord. **Option C is the diameter of the small circle; the chord is strictly shorter.**

5. Three circles of radii 3, 4 and 5 are pairwise externally tangent (each pair has exactly one common point). What is the area of the triangle formed by their three centres?

- A) $12\sqrt{5}$
- B) 12
- C) 84
- D) $3\sqrt{5}$

Answer: A) $12\sqrt{5}$

Method: The centres form a triangle with sides $3 + 4 = 7$, $3 + 5 = 8$, $4 + 5 = 9$. Heron: $s = \frac{7+8+9}{2} = 12$, area $= \sqrt{12 \cdot 5 \cdot 4 \cdot 3} = \sqrt{720} = 12\sqrt{5}$.

Investigate further: Co-ordinate check: $(0, 0)$ and $(7, 0)$ fixed, third centre 9 from the first and 8 from the second: $x = \frac{33}{7}$, $y = \frac{24\sqrt{5}}{7}$, area $= \frac{1}{2} \cdot 7 \cdot \frac{24\sqrt{5}}{7} = 12\sqrt{5}$ ✓. **The side length is the sum of the two radii, never the difference.**

Top 5 Patterns — Worksheet #3

- Classify two circles by comparing d with $r_1 + r_2$ and $|r_1 - r_2|$.** $d > r_1 + r_2$ (4 tangents), $d = r_1 + r_2$ (3), overlap (2), $d = |r_1 - r_2|$ (1), containment (0). (see A1, A2, A6, B1, B6)
- Radical axis = the two equations subtracted.** The $x^2 + y^2$ terms vanish in one line; the result is the common chord for intersecting circles and the equal-power line in general. (see A3, B2, C5, D1)
- Common chord = radical axis + half-chord formula.** Find the radical axis, take its distance d from a centre, then $2\sqrt{r^2 - d^2}$. (see A5, B8, C7, D2, D4)
- Orthogonality either way:** $d^2 = r_1^2 + r_2^2$ or $2g_1g_2 + 2f_1f_2 = c_1 + c_2$. The radii make a right triangle against the centre distance. (see A4, A8, B3, B7, C2, D3)
- A circle tangent to both axes lives at $(\pm r, \pm r)$ with constant r^2 .** Powers of r are found from a point condition; there can be two solutions. (see A7, B4, B9, C6)

Common Traps — Worksheet #3

- Miscounting the common-tangent table.** Externally tangent means exactly 3 common tangents (diameter line plus two direct); crossing circles have 2, disjoint circles 4. One boundary-off-by-one changes the whole count. (see A1, A2, A6, A10, B1, B6)
- Mangling the subtraction sign on the radical axis.** It is the difference of the two constants and linear terms set to zero; flip a sign and you get the next option in C5. (see A3, B2, C5)
- Wrong pairing in the orthogonality test.** The condition binds $g_1g_2 + f_1f_2$ against $(c_1 + c_2)/2$ — pair g with g , f with f . (see A8, B3, B7, D3)
- Imagining the radical axis is a circle.** Equal powers trace a straight line, not a curve, even between disjoint circles. (see D1, B5, B10)
- Forgetting the triangle of centres on tangent circles.** For pairwise tangent circles each centre distance is the sum of the two radii; subtracting instead of adding breaks Heron immediately. (see D5, A10, A1)

“Two circles are solved by one subtraction: locate, subtract, then compare with the sum and difference of the radii.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths